Matrices, Determinants and Systems of linear equations

Operations of determinants:

1) $L_i \leftrightarrow L_j$ (swap)

2) $L_i \leftrightarrow \alpha L_i$ (multiplying by a scalar)

3) $L_i \leftrightarrow \alpha L_j + L_i$ (mult. a diff. line then add)

ex.1) $D = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & 1 \\ -1 & 4 & -1 & 0 \\ 0 & 2 & 1 & 1 \end{vmatrix} \overset{\substack{L_2-2L_1 \\ L_3+L_1}}{=\!=} \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & -5 \\ 0 & 6 & -2 & 3 \\ 0 & 2 & 1 & 1 \end{vmatrix}$

$= 1 \cdot \begin{vmatrix} -5 & 5 & -5 \\ 6 & -2 & 3 \\ 2 & 1 & 1 \end{vmatrix} \cdot (-1)^{1+1} = \begin{vmatrix} -5 & 5 & -5 \\ 6 & -2 & 3 \\ 2 & 1 & 1 \end{vmatrix} = 5 \cdot \begin{vmatrix} -1 & 1 & -1 \\ 6 & -2 & 3 \\ 2 & 1 & 1 \end{vmatrix}$

$\overset{\substack{C_1+C_2 \\ C_1+C_3}}{=\!=} 5 \cdot \begin{vmatrix} 0 & 1 & 0 \\ 4 & -2 & 1 \\ 3 & 3 & 2 \end{vmatrix} = 1 \cdot (-1)^{1+2} \cdot \begin{vmatrix} 4 & 1 \\ 3 & 2 \end{vmatrix} \cdot 5 = -5 \cdot 5 = -25$

2) $D = \begin{vmatrix} -2-a & -1 & 1 \\ 5 & -1-a & 4 \\ 5 & 1 & 2-a \end{vmatrix} \overset{C_3+C_1}{=\!=} \begin{vmatrix} -2-a & -1 & 0 \\ 5 & -1-a & 3-a \\ 5 & 1 & 3-a \end{vmatrix}$

$= (3-a) \begin{vmatrix} -2-a & -1 & 0 \\ 5 & -1-a & 1 \\ 5 & 1 & 1 \end{vmatrix} \overset{\substack{(3-a). \\ L_2-L_3}}{=\!=} \begin{vmatrix} -2-a & -1 & 0 \\ 0 & -2-a & 0 \\ 5 & 1 & 1 \end{vmatrix}$

$$= (3-a)\cdot 1\cdot (-1)^{3+3}\cdot \begin{vmatrix} -2-a & 1 \\ 0 & -2-a \end{vmatrix} = (a-3)(a+2)\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix}$$

$$= (a-3)(a+2)$$

$$D(a)=0 \iff (a-3)(a+2)=0 \Rightarrow a_1 = 3$$
$$a_2 = -2$$

3) $A = \begin{pmatrix} 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix} \overset{L_3-L_1}{=} \begin{pmatrix} 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 2 & -1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$

$\overset{\substack{L_3-L_2 \\ L_4-L_2}}{\sim} \begin{pmatrix} 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \operatorname{rank}(A) = 3$

4) $B = \begin{pmatrix} 2 & 1 & 3 & -1 \\ 3 & -1 & 2 & 0 \\ 1 & 3 & 4 & -2 \\ 4 & -3 & 1 & 1 \end{pmatrix} \overset{L_3 \leftrightarrow L_4}{\sim} \begin{pmatrix} 1 & 3 & 4 & -2 \\ 3 & -1 & 2 & 0 \\ 2 & 1 & 3 & -1 \\ 4 & -3 & 1 & 1 \end{pmatrix}$

$\overset{\substack{L_2-3L_1 \\ L_3-2L_1 \\ L_4-4L_1}}{\sim} \begin{pmatrix} 1 & 3 & 4 & -2 \\ 0 & -10 & -10 & 6 \\ 0 & -5 & -5 & 3 \\ 0 & -15 & -15 & 9 \end{pmatrix} \overset{L_2 \leftrightarrow L_3}{\sim} \begin{pmatrix} 1 & 3 & 4 & -2 \\ 0 & -5 & -5 & 3 \\ 0 & -10 & -10 & 6 \\ 0 & -15 & -15 & 9 \end{pmatrix}$

$\overset{\substack{L_3-2L_2 \\ L_4-3L_2}}{\sim} \begin{pmatrix} 1 & 3 & 4 & -2 \\ 0 & -5 & -5 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \operatorname{rank}(B) = 2$

$$[A|I_4] = \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 2 & -1 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} L_3 + L_1 \\ L_4 - 2L_1 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -3 & -2 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} \\ \\ \\ L_4 + L_2 \end{array} \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -4 & -2 & 1 & 1 & 1 \end{array}\right]$$

$$\begin{array}{l} L_4 \leftrightarrow L_3 \\ \\ \\ \\ \\ \end{array} \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & -4 & -2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array}\right] \begin{array}{l} L_3 + 4L_4 \\ L_2 + L_3 \\ L_1 - 2L_2 \\ \\ \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & -1 & 0 & -2 & 0 \\ 0 & -1 & 3 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 & 2 & 1 & 4 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array}\right] \begin{array}{l} \\ \\ \\ \\ \end{array} \begin{array}{l} \text{Factor } 2 \\ \\ \\ \\ \end{array} \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & -1 & 0 & -2 & 0 \\ 0 & -1 & 3 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & \frac{1}{2} & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array}\right]$$

$$\begin{array}{l} \\ L_2 - 3L_3 \\ \\ \\ \\ \end{array} \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & -1 & 0 & -2 & 0 \\ 0 & -1 & 0 & 0 & -2 & -\frac{1}{2} & -5 & -\frac{3}{2} \\ 0 & 0 & 1 & 0 & 1 & \frac{1}{2} & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array}\right] \begin{array}{l} \\ \\ \text{Factor}(-1) \\ \\ \\ \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & -1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 0 & 2 & \frac{1}{2} & 5 & \frac{3}{2} \\ 0 & 0 & 1 & 0 & 1 & \frac{1}{2} & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array}\right] \begin{array}{l} \\ L_1 + L_2 \\ \\ \\ \\ \end{array} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & \frac{1}{2} & 3 & \frac{3}{2} \\ 0 & 1 & 0 & 0 & 2 & \frac{1}{2} & 5 & \frac{3}{2} \\ 0 & 0 & 1 & 0 & 1 & \frac{1}{2} & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array}\right]$$

I_4 $\qquad$ A^{-1}